

Session 4 · Building a Company Screening Engine

**A program that screens 16 companies on live SEC filings
and explains every company that passes the screen.**

Plan: concepts 15' · live demo 22' · your lab 30' · debrief 8'

By the end of Session 4 you can

1. **Draw** the workflow pattern and defend the code-vs-model split
2. **Fetch** three years of revenue and profit for any ticker from its filings
3. **Screen** 16 companies on criteria you control
4. **Audit** the model's explanations: every number traced to your data
5. **Train** a revenue forecaster and measure its error before trusting it
6. **Express** the whole pipeline as a LangGraph graph

What we are doing, and why

- **We are building a company screening engine:** fetch fundamentals, apply tests you control, explain each survivor in audited prose.
- **The machinery is the point:** you assemble the fetch, the filter and the audit — reusable on any companies, any criteria.
- **A workflow is a plan the model cannot change:** code fetches and filters; the model explains; code checks the explanations; you approve.
- **The finished pipeline is a LangGraph graph** — the professional form for workflows like this.

The storyline of this session

Apple's raw filing record
↓
a fetcher for any ticker
↓
the screen
↓
rationales, audited
↓
a revenue forecaster
↓
the pipeline as a graph

revenue and profit, straight from EDGAR
you assemble it; 16 companies pass through
two tests you control – does Apple pass?
2 sentences per survivor, every number traced
next year's revenue, with its measured error
LangGraph: the same steps, drawn and run

A workflow is a plan the model cannot change

INPUT → RETRIEVE → STRUCTURE → REASON → VALIDATE → HUMAN
(code) (code) (model) (code) (you)

- **Code does the arithmetic:** pandas is deterministic; the model is not
- **The model reasons once,** in the middle, about a table it is *given*
- **Code audits the prose,** and a human approves: nothing is published without explicit sign-off

The data source: SEC EDGAR

- **EDGAR** (Electronic Data Gathering, Analysis and Retrieval) is the U.S. Securities and Exchange Commission's public filing database
- **What it contains:** every 10-K, 10-Q and 8-K since the mid-1990s, machine readable, free of charge, no license required
- **What it does not contain:** market prices and analyst estimates — filings are not market data

Three complications you will meet in the lab, because **companies tag their own accounts**:

1. The same item **changes names** across years
2. Foreign filers report in **local currency**
3. Most pharma reports **no operating income line at all**

Demo: the workflow end to end

A company memo built live from the filings. Two things to observe:

- **The data gaps:** the memo lists what it could *not* know from its inputs — the blind spots are declared, not hidden
- **A planted error:** one figure in the memo is altered by hand, and the audit finds it

```
audit: every figure in the prose → checked against inputs → the altered  
figure is flagged
```

The libraries in this session

Library	Role here	Standing
<code>requests</code>	fetches SEC filings over HTTP	the standard HTTP library
<code>pandas</code>	the deterministic screen	the industry standard for data work
<code>numpy</code>	trains the least-squares forecaster	the numerical foundation of Python
<code>pydantic</code>	typed, validated model output	the industry standard for validation
LangGraph	the pipeline as an explicit graph	a leading framework for AI workflows, from the LangChain ecosystem

How the labs work

Every exercise sits between two markers. **You fill the gaps. Nothing else changes.**

```
### START CODE HERE ###  
mask = (df[None] >= min_growth) & (df[None] >= min_margin)  
### END CODE HERE ###
```

- **None** → replace with the correct column, value or variable
- **[QUESTION IN CAPITALS]** → replace with the text the bracket asks for
- **Everything else is given.** Do not rewrite it.
- Then run the  **check cell** directly below. Green means correct: continue.

Stuck for two minutes? Select the lines, press **Option+K** (**Alt+K** on Windows), and ask the ***** panel.

Your lab · notebook 04 · 30 minutes

A screening engine: Apple's peer universe + pharma + industrials, live from EDGAR

- **Exercise 1:** fetch 3 years of fundamentals for 16 companies, tolerating failed tickers
- **Exercise 2:** the filter — growth $\geq 8\%$ and margin $\geq 10\%$, **pandas** decides which companies pass, never the model. Note what happens to Apple
- **Exercise 3:** one grounded rationale per survivor + numeric audit
- **Exercise 4:** train a revenue forecaster and measure its error on a held-out year
- **Exercise 5:** wire the same pipeline as a LangGraph graph — the graph prints its own diagram

Two questions for the debrief: why net margin, not operating margin?

Key takeaway

The value of this session is repeatability: a screen you can rerun any morning, with criteria you control and prose your code audits.

Next session: the model plans its own steps, under controls you define — and your finished tool goes on GitHub.